

Lecture 11: **Causal Inference**

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Advanced Text Analysis

Concept(s): Pretraining and Fine-tuning

- **Pre-train** a model on a task where we have lots of data
 - *e.g., we have lots and lots of internet data*
- **Fine-tune** the model on your downstream task
 - *e.g., a specific, curated dataset that you're studying => sentiment analysis*
 - You can use the exact same dataset that you would have used to train a classifier from scratch



Lifecycle of a BERT model

A BERT model consists of ~400M of numbers. ← parameters
Where do they come from, and when do they change?

Requires more time and computational resources

Pre-training

(unsupervised!)

Gradually improve parameters by looping over large datasets



WIKIPEDIA
The Free Encyclopedia

Fine-tuning

Gradually improve parameters *for a new purpose* by looping over smaller datasets

Encode new docs

Take new inputs, produce outputs, but make *no change* to model parameters

Production use

Take new inputs, produce outputs, make *no change* to parameters

word
← similarity
example

genre
classification
example

Adapted from [Antoniak and Walsh, “BERT for Social Sciences and Humanities”, Tutorial at ICWSM](#)

BERT vs. Generative Large Language Models (LLMs)

- What is the difference between the BERT style models vs. ChatGPT?

Major Large Language Models (LLMs)

ranked by capabilities, sized by billion parameters used for training

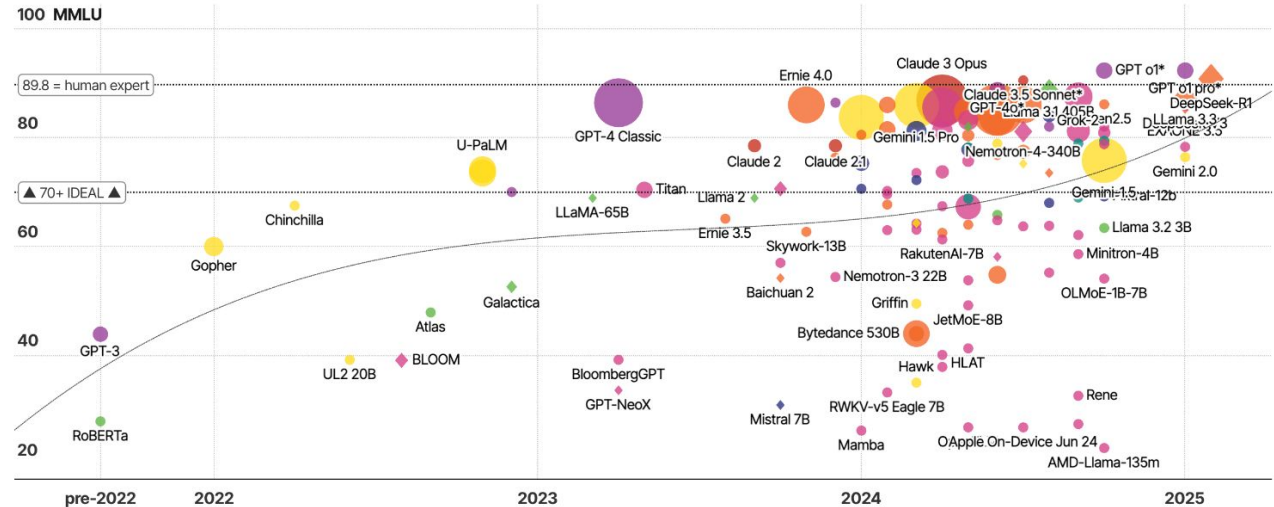
CLICK LEGEND ITEMS TO FILTER

Parameters (Bn)  open access

anthropic chinese google meta microsoft mistral openAI other

🔍 search...

show only: all

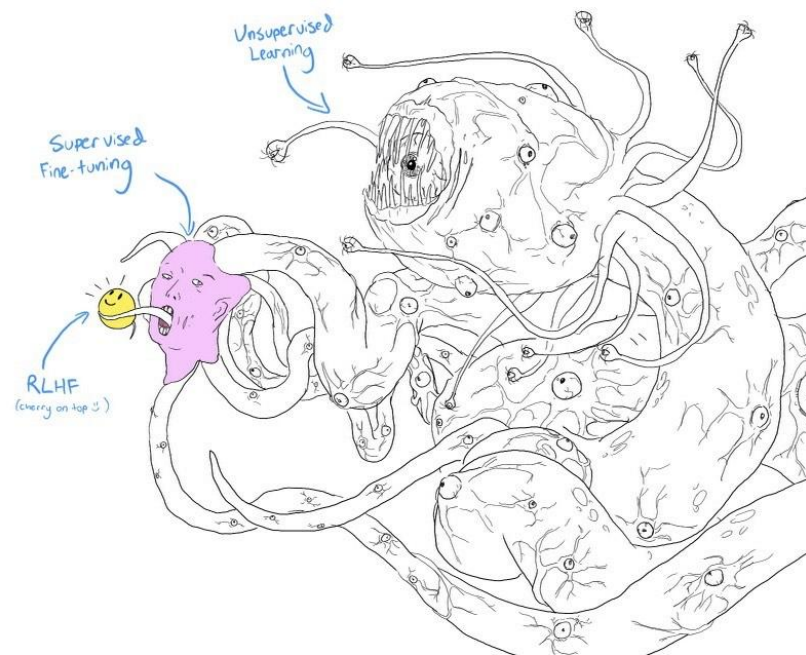
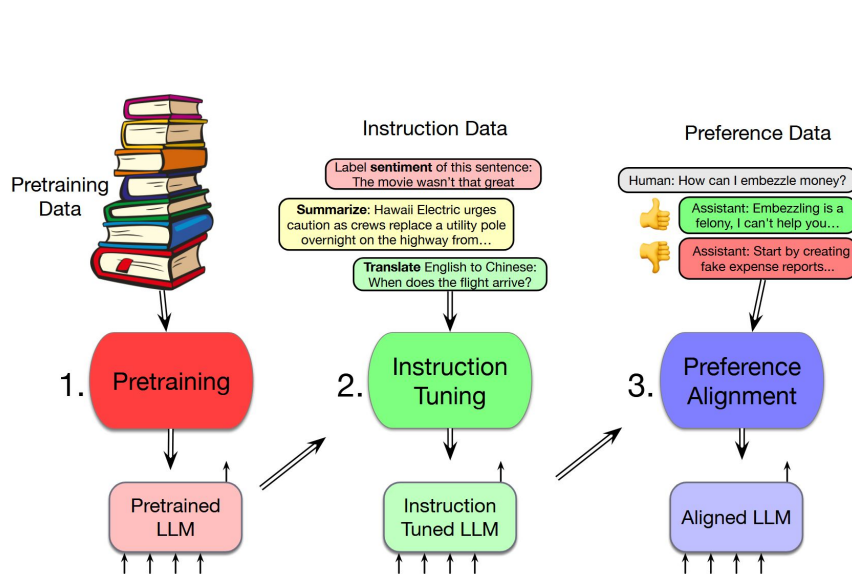


David McCandless, Tom Evans, Paul Barton
Informationisbeautiful // Jan 2024

MMU = benchmark for measuring LLM capabilities
* = parameters undisclosed // source: LifeArchitect // data

<https://www.sistrix.de/frag-sistrix/ai-grundlagen/large-language-model/>

Generative LLMs which we can prompt

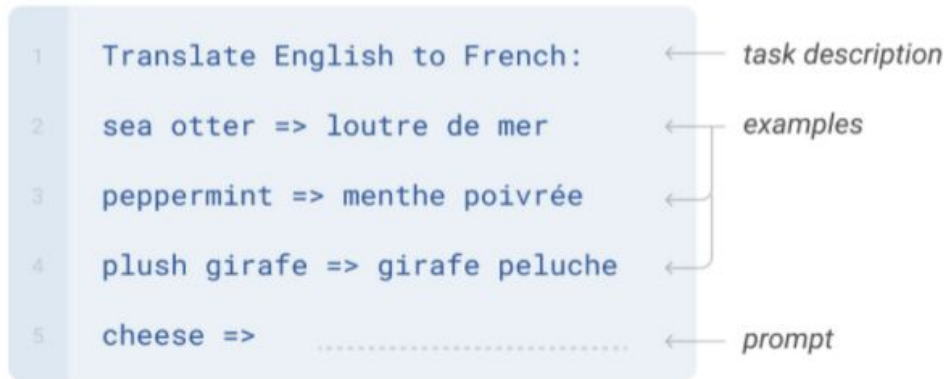


Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing <https://web.stanford.edu/~jurafsky/slp3>

Shoggoth with Smiley Face. Courtesy of twitter.com/anthrupad

Prompt based labeling

- Prompt: a text string that a user issues to a language model to get the model to do something useful by conditional generation
- Zero-shot: only instructions, no examples
- Few-shot: instructions + (a few) examples
- More advanced prompting strategies:
 - Chain-of-thought reasoning (COT)
 - Self-Consistency



<https://huggingface.co/blog/few-shot-learning-gpt-neo-and-inference-api>

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. **X**

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 **X**

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

<https://www.promptingguide.ai/techniques/cot>

Today: Causal Inference

Taking stock of what we've seen so far

So far we have studied DESCRIPTIVE questions:

- How does news media represent the immigration crisis?
- What topics dominate posts about elections?
- What is the sentiment toward ICE on Bluesky?

Today: CAUSAL questions

- Does news framing **change** attitudes toward immigrants?
- Did banning a community **reduce** hate speech on the platform?
- How does exposure to negative posts **affect** users' own emotions?

What is causal inference?

- Causal inference: the process of determining *whether* (and *how much*) one variable **influences** another variable
 - Treatment (T): the cause we care about (e.g., exposure to a post, a platform ban)
 - Outcome (Y): the effect we measure (e.g., engagement, hate speech use)
- We want to estimate the causal effect of T on Y
 - Not just: are T and Y correlated?
 - But: would Y change if we changed T, holding everything else equal?

The fundamental question:

$$E[Y \mid \text{do}(T=1)] - E[Y \mid \text{do}(T=0)]$$

"What would happen on average if we forced T to be 1 vs. forced T to be 0?"

Causal vs. descriptive questions

Same data, very different questions:

DESCRIPTIVE

What share of posts mention vaccines?

Are users in echo chambers more polarized?

Which subreddits have the most toxic comments?

How does engagement vary across topics?

CAUSAL

Does seeing pro-vaccine content increase intent to vaccinate?

Does joining an echo chamber make a user more polarized?

Did banning r/X reduce toxicity for its former members?

Does moral-emotional language cause higher engagement?

Watch the verbs: "is associated with" vs "causes" / "affects" / "changes"

Why do causal inference?

- Description tells us what the world looks like. Causation tells us what to do about it.
 - **Policy**: should the platform ban this community? Does that actually reduce harm?
 - **Design**: does adding a friction step before sharing reduce misinformation spread?
 - **Theory**: do the **mechanisms** we hypothesize about online behavior actually hold?
- Many social science questions are inherently causal
 - Effects of media exposure, of platform policies, of online interventions
- Without causal claims, we cannot guide interventions or test theories

Why is causal inference hard?

The fundamental problem of causal inference ([Holland, 1986](#)):

We can never observe the same unit both **treated** AND **untreated** at the same time.

FACTUAL (observed)

Alice was exposed to misinformation.

She shared 4 posts that day.

$Y(T=1) = 4$ ✓ observed

COUNTERFACTUAL (unobservable)

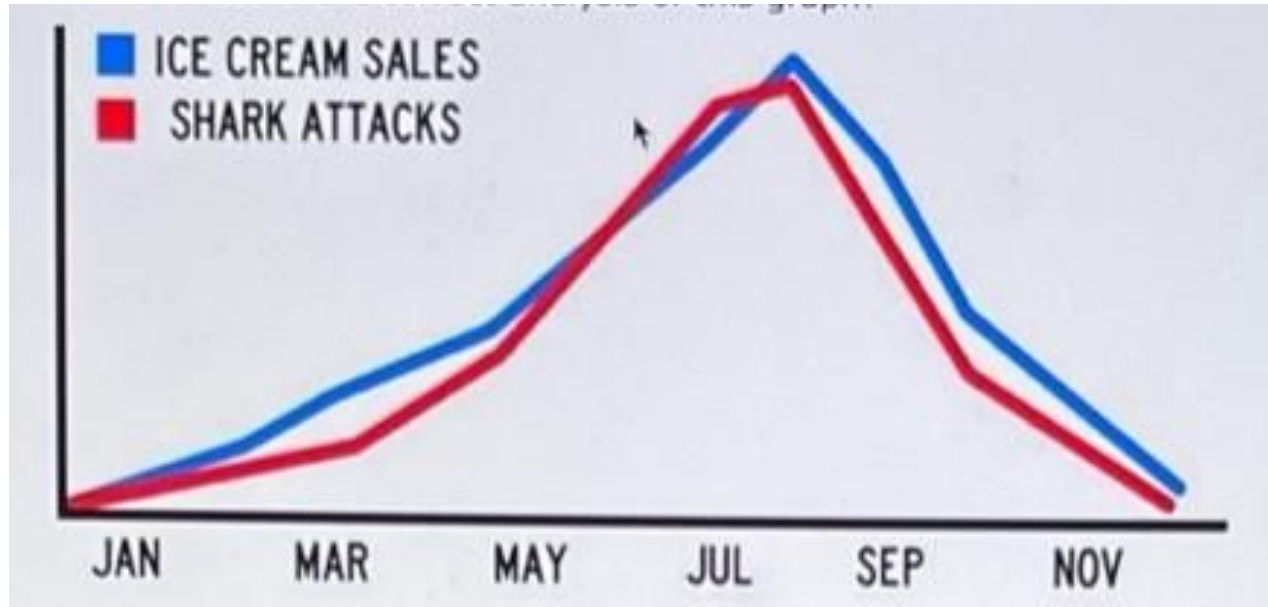
If Alice had NOT been exposed...

How many posts would she have shared?

$Y(T=0) = ?$ ✗ never observed

*Causal inference = the science of estimating the missing **counterfactual**.*

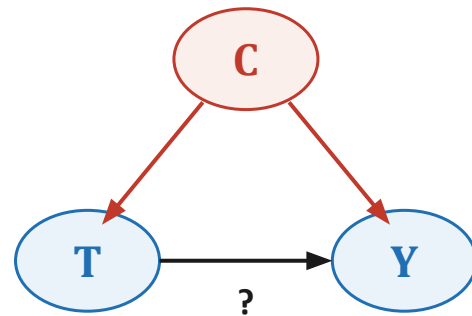
Why is causal inference hard?



<https://www.chegg.com/homework-help/questions-and-answers/graph-shows-ice-cream-sales-increase-shark-attacks-also-increase-know-ice-cream-isn-t-good-q85332558>

Why is causal inference hard? Confounding

- A confounder is a variable that influences BOTH treatment and outcome
 - Creates a **spurious association** between T and Y
- Example: heavy social media users post more outrage
 - Does outrage → engagement, or do engaged people post outrage?
 - Or: a third factor (personality, news cycle) drives both?
- On social media, confounders are everywhere:
 - User-level: personality, demographics, prior beliefs, time online
 - Platform-level: algorithm, network position, time-of-day
 - Time-varying: news events, trends, seasonality

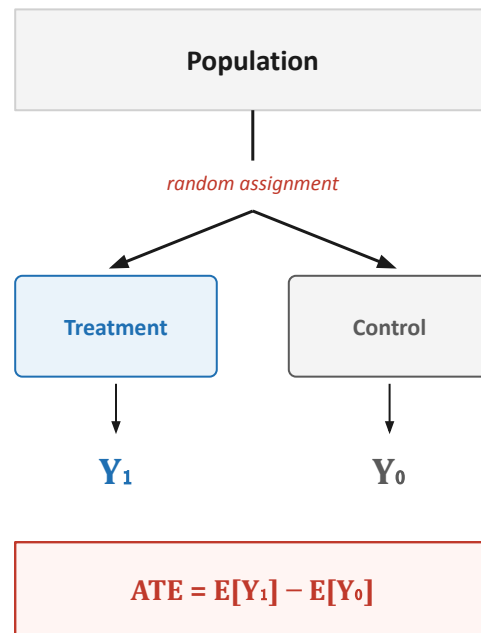


Confounder C creates a **back-door** path between T and Y

How do we establish causality?

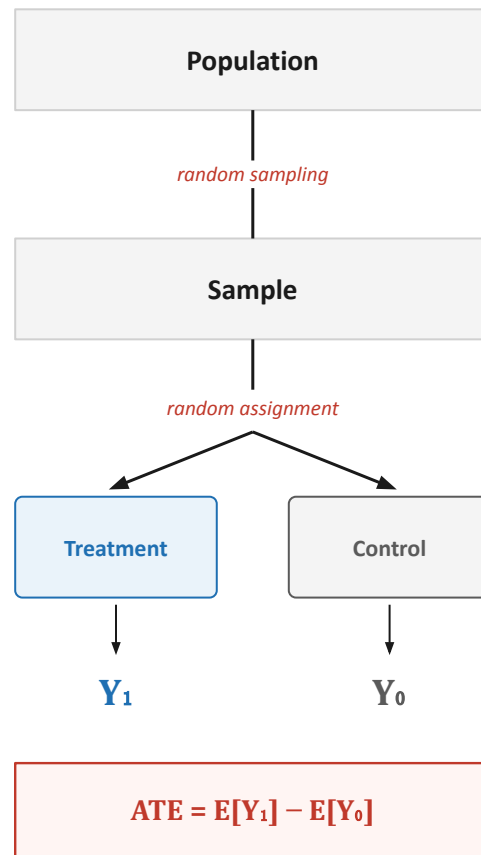
Experiments: the gold standard

- In a **Randomized Controlled Trial (RCT)**, we randomly assign units to treatment or control
 - Random assignment makes groups equivalent in expectation
 - All confounders—observed or unobserved—are balanced across groups
- The simple difference in means estimates the **average treatment effect (ATE)**
- By intervening on T, we break the path from any confounder C to T
 - Now, any difference in Y must be due to T



Experiments: the gold standard

- It might not be possible to work with the entire population, especially in experiments
- So in addition to random assignment, we first need **random sampling**
- Random sampling allows us to obtain a sample **representative** of the population.
 - Therefore, results of the study can be generalized to the population.



	Random assignment	No random assignment	
Random sampling	Causal inference, generalized to the whole population.	No causal inference, correlation statement generalized to the whole population.	Conclusions generalized to population.
No random sampling	Causal inference, only for the sample.	No causal inference, correlation statement only for the sample.	Conclusions not generalized to population.
	Causation	Correlation	

Mine Cetinkaya-Rundel's Statistics 101 Lecture:

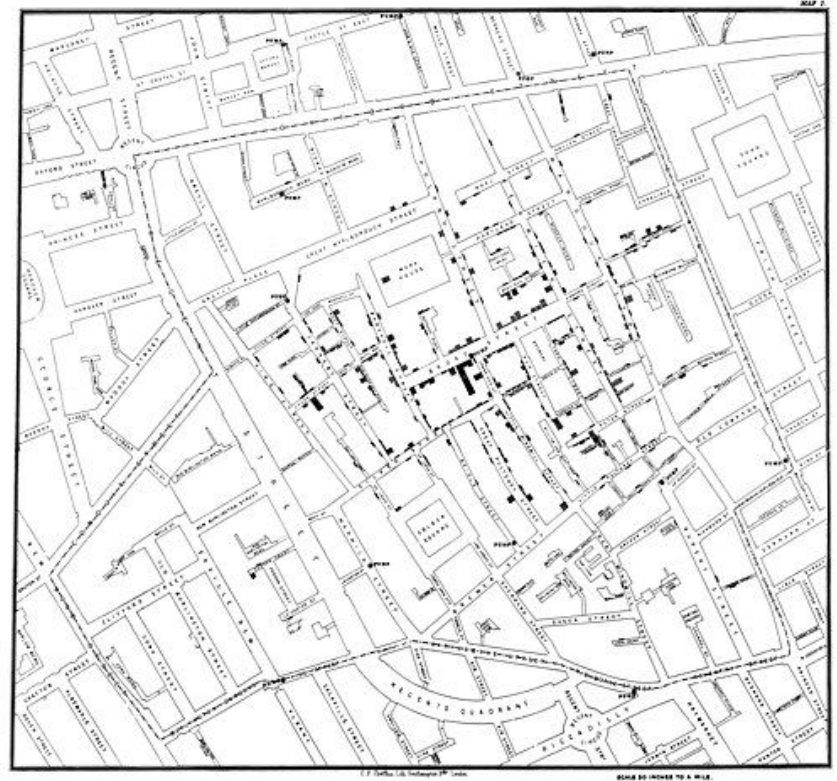
https://www2.stat.duke.edu/courses/Fall12/sta101.001/resources/lecturettes/random_sample_assignment.pdf

When experiments aren't possible: observational data

- We often cannot (or should not) run experiments
 - Ethics: we can't randomly expose people to misinformation
 - Feasibility: we don't control the platform's algorithm
 - History: many questions are about events that already happened
- With observational data, **treatment is NOT randomly assigned**
 - Users self-select into being exposed to certain content
 - This selection is almost always confounded
- Two main strategies for observational causal inference:
 - Natural experiments — find a setting where treatment is "as good as random"
 - Adjustment-based methods — measure and condition on confounders (matching, regression, weighting)

Natural experiments

- A natural experiment is an event in the real world that approximates random assignment
 - Some users are "treated" by chance, history, or policy
 - Not treated by their own choice
- Examples:
 - **Cholera epidemic in London in the 1850s:** which households were most likely to get cholera and why?
 - **Vietnam war draft in the U.S in the 1950s and 1960s:** What role does military service play in people's lifetime earnings?

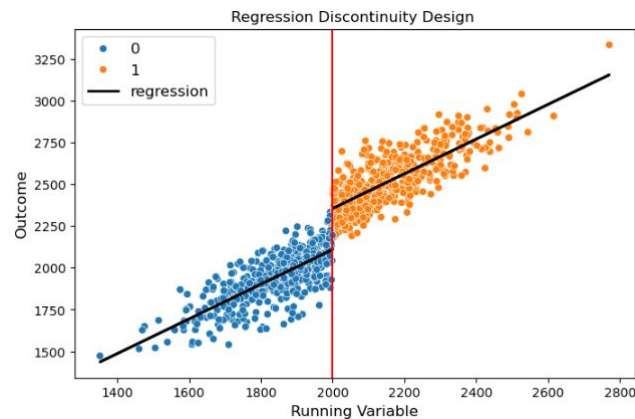
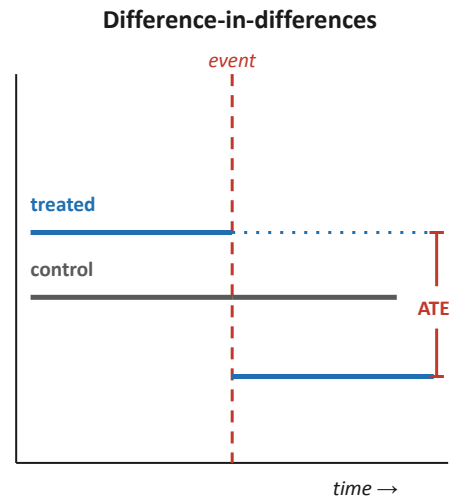


John Snow's map showing the clustering of cholera cases in Soho during the London epidemic of 1854

https://en.wikipedia.org/wiki/Natural_experiment

Natural experiments: Common designs

- Difference-in-differences (DiD): compare change-in-Y for treated vs control (untreated) groups
- Regression discontinuity (RD): compare units just above vs just below a threshold



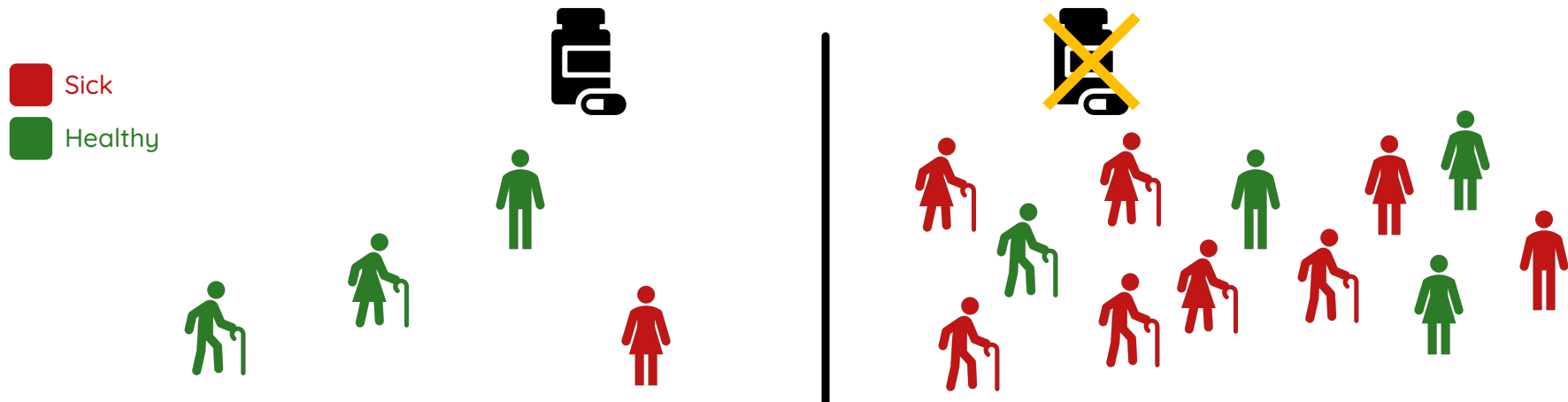
<https://medium.com/@arun.subram456/causal-inference-regression-discontinuity-design-338f0f0b5f31>

Another Approach: Matching-based studies

- Idea: for every treated unit, find an *untreated* unit that looks the same on observed covariates (i.e., related variables)
 - Compare outcomes within these matched pairs
 - Mimics the balance that random assignment gives us
- Useful when you have rich user-level data and no natural experiment

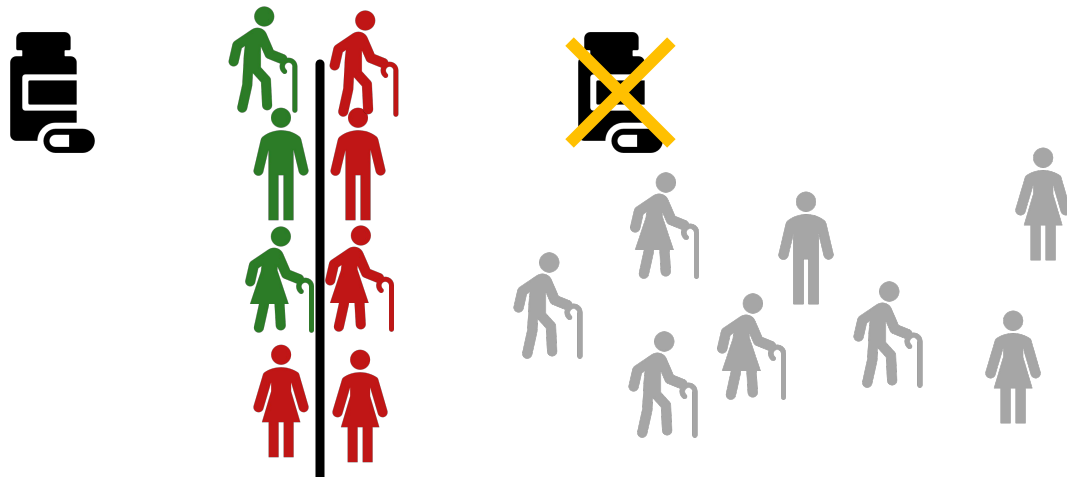
~Weight Models: Matching

- A very popular and intuitive method for adjustment:
 - For each treated unit find the control most similar to it.



~Weight Models: Matching

- A very popular and intuitive method for adjustment:
 - For each treated unit find the control most similar to it.



~Weight Models: Matching

- Using matching as
 - a potential outcome prediction model



Matching-based studies

- Typical steps in matching approach:
 - Define treatment, outcome, and a set of covariates that confound them
 - Match each treated user to one (or more) similar untreated user(s)
 - Check covariate balance after matching
 - Estimate the effect on the matched sample, i.e., the Average Treatment Effect
- Matching is **critical** when you say you want to differentiate A from non-A *because of A*
 - A = treatment
 - E.g., what is the difference between banned users and non-banned users because the former were banned?

There are many different matching approaches: some examples:

Exact matching	Match on identical values of every covariate	+ Simple, transparent - Only works with few, discrete covariates (curse of dimensionality)
Propensity score matching	Model $P(T=1 \mid \text{covariates})$, then match on this single score	+ Reduces many covariates to one number - Quality depends on the model; only balances on observed covariates
Coarsened exact matching	Coarsen covariates into bins, then exact match on the bins	+ Easy, guarantees balance on coarsened variables - Choice of bins affects results
Stuart-style / Mahalanobis	Match using a multivariate distance over the covariate space	+ Accounts for covariate correlations - Sensitive to distance metric and outliers

Deeper dive: King, Gary, et al. "[Comparative effectiveness of matching methods for causal inference.](#)" *Unpublished manuscript, Institute for Quantitative Social Science, Harvard University, Cambridge, MA* (2011).

Causal Inference with Social Media Data

Causal Inference with Social Media Data

But, first please fill out the official course feedback!

Experiments on social media platforms

- Researchers (sometimes) collaborate with platforms to run real experiments
 - Treatment: alter what some users see (feed content, notifications, ranking)
 - Outcome: measure resulting behavior (posting, engagement, sentiment)
- Example: [Kramer et al.](#) manipulated News Feeds for 600K+ Facebook users
 - Reduced positive (or negative) emotional content in some users' feeds
 - Found small but significant emotional contagion effects in users' own posts
- Famous and contentious; also raised major ethical concerns about consent

Experimental evidence of massive-scale emotional contagion through social networks

Adam D. I. Kramer^{a,1}, Jamie E. Guillory^{b,2}, and Jeffrey T. Hancock^{b,c}

^aCore Data Science Team, Facebook, Inc., Menlo Park, CA 94025; and Departments of ^bCommunication and ^cInformation Science, Cornell University, Ithaca, NY 14853

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 25, 2014 (received for review October 23, 2013)

Emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. Emotional contagion is well established in laboratory experiments, with people transferring positive and negative emotions to others. Data from a large real-world social network, collected over a 20-y period suggests that longer-lasting moods (e.g., depression, happiness) can be transferred through networks [Fowler JH, Christakis NA (2008) *BMJ* 337:a2338], although the results are controversial. In an experiment with people who use Facebook, we test whether emotional contagion occurs outside of in-person interaction between individuals by reducing the amount of emotional content in the News Feed. When positive expressions were reduced, people produced fewer positive posts and more negative posts; when negative expressions were reduced, the opposite pattern occurred. These results indicate that emotions expressed by others on Facebook influence our own emotions, constituting experimental evidence for massive-scale contagion via social networks. This work also suggests that, in contrast to prevailing assumptions, in-person interaction and nonverbal cues are not strictly necessary for emotional contagion, and that the observation of others' positive experiences constitutes a positive experience for people.

computer-mediated communication | social media | big data

Emotional states can be transferred to others via emotional contagion, leading them to experience the same emotions as those around them. Emotional contagion is well established in laboratory experiments (1), in which people transfer positive and negative moods and emotions to others. Similarly, data from a large, real-world social network collected over a 20-y period suggests that longer-lasting moods (e.g., depression, happiness) can be transferred through networks as well (2, 3).

The interpretation of this network effect as contagion of mood has come under scrutiny due to the study's correlational nature, including concerns over misspecification of contextual variables or failure to account for shared experiences (4, 5), raising important questions regarding contagion processes in networks. An experimental approach can address this scrutiny directly; however, methods used in controlled experiments have been criticized for examining emotions after social interactions. Interacting with a happy person is pleasant (and an unhappy person, unpleasant). As such, contagion may result from experiencing an interaction rather than exposure to a partner's emotion. Prior studies have also failed to address whether nonverbal cues are

demonstrated that (i) emotional contagion occurs via text-based computer-mediated communication (7); (ii) contagion of psychological and physiological qualities has been suggested based on correlational data for social networks generally (7, 8); and (iii) people's emotional expressions on Facebook predict friends' emotional expressions, even days later (7) (although some shared experiences may in fact last several days). To date, however, there is no experimental evidence that emotions or moods are contagious in the absence of direct interaction between experimenter and target.

On Facebook, people frequently express emotions, which are later seen by their friends via Facebook's "News Feed" product (8). Because people's friends frequently produce much more content than one person can view, the News Feed filters posts, stories, and activities undertaken by friends. News Feed is the primary manner by which people see content that friends share. Which content is shown or omitted in the News Feed is determined via a ranking algorithm that Facebook continually develops and tests in the interest of showing viewers the content they will find most relevant and engaging. One such test is reported in this study: A test of whether posts with emotional content are more engaging.

The experiment manipulated the extent to which people ($N = 689,003$) were exposed to emotional expressions in their News Feed. This tested whether exposure to emotions led people to change their own posting behaviors, in particular whether exposure to emotional content led people to post content that was consistent with the exposure—thereby testing whether exposure to verbal affective expressions leads to similar verbal expressions, a form of emotional contagion. People who viewed Facebook in English were qualified for selective assignment to the experiment. Two parallel experiments were conducted for positive and negative emotion: One in which exposure to friends' positive emotional content in their News Feed was reduced, and one in which exposure to negative emotional content in their News Feed was reduced. In these conditions, when a person loaded their News Feed, posts that contained emotional content of the relevant significance, each emotional post had between a 10% and

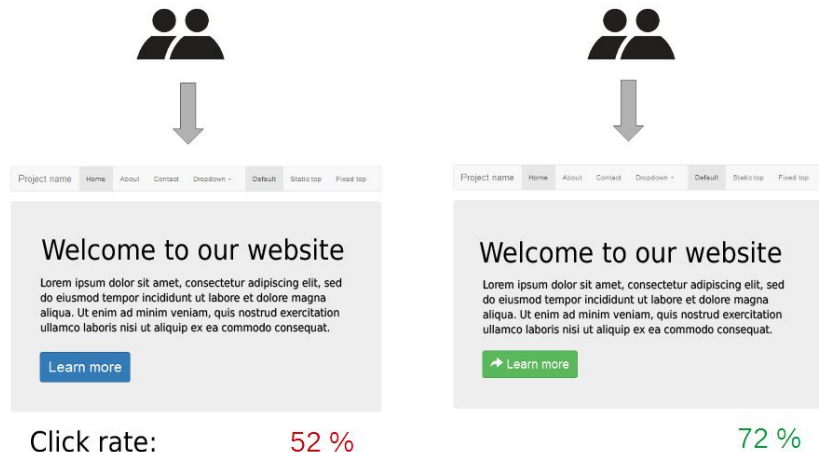
Significance

We show, via a massive ($N = 689,003$) experiment on Facebook, that emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. We provide experimental evidence that emotional contagion occurs without direct interaction be-



Experiments on social media platforms: A/B tests

- A/B testing: industry term for a randomized controlled trial run on a live platform
 - Users are randomly split into group A (current experience) and group B (new variant)
 - The platform measures an outcome metric (clicks, time-on-site, retention) for each group
- Example of what gets A/B tested on social media
 - Feed ranking changes, notification timing, button placement, recommendation models
 - Even policy interventions: friction prompts, content warnings, "are you sure?" dialogs



https://en.wikipedia.org/wiki/A/B_testing

Experiments on social media platforms

- All social media experiments are not necessarily unethical or infeasible
- Create own experimental platform; similar to a lab study
 - High levels of control
 - Low external validity → might not generalize to real-world settings

Mahajan, K., Roy Choudhury, S., Levens, S., Gallicano, T., & Shaikh, S. (2021, March). [Community connect: A mock social media platform to study online behavior](#). In *Proceedings of the 14th ACM international conference on web search and data mining* (pp. 1073-1076).

Community Connect: A Mock Social Media Platform to Study Online Behavior

Khyati Mahajan, Sourav Roy Choudhury, Sara Levens, Tiffany Gallicano, Samira Shaikh
University of North Carolina at Charlotte
[kmahaja2,sroychou,slevens,tgallica,sshaikh2]@unc.edu

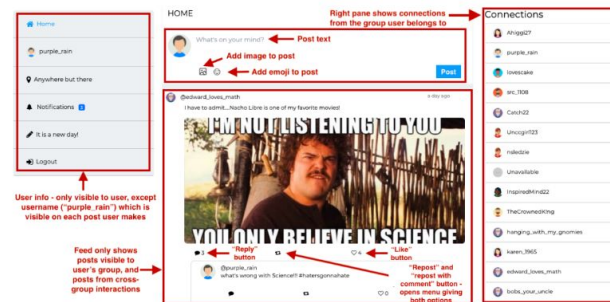


Figure 1: Interface of the Community Connect social networking platform and key features that are supported

ABSTRACT

We present Community Connect, a custom social media platform for conducting controlled experiments of human behavior. The key distinguishing factor of Community Connect is the ability to control the visibility of user posts based on the groups they belong to, allowing careful and controlled investigation into how information propagates through a social network. We release this platform as a resource to the broader community, to facilitate research on data collected through controlled experiments on social networks.

CCS CONCEPTS

ACM Reference Format:

Khyati Mahajan, Sourav Roy Choudhury, Sara Levens, Tiffany Gallicano, Samira Shaikh. 2021. Community Connect: A Mock Social Media Platform to Study Online Behavior. In *Proceedings of the Fourteenth ACM International Conference on Web Search and Data Mining (WSDM '21)*, March 8–12, 2021, Virtual Event, Israel. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3437963.3441698>

1 INTRODUCTION

Social media platforms, including Twitter, Instagram and Facebook, have become extremely popular for studying patterns of human

Experiments on social media platforms

- All social media experiments are not necessarily unethical or infeasible
- Create own experimental platform; similar to a lab study
 - High levels of control
 - Low external validity → might not generalize to real-world settings
- Alternative: Field experiments
 - Experiments carried out in real-world settings
 - High external validity, less control

Munger, K. (2017). [Tweetment effects on the tweeted: Experimentally reducing racist harassment](#). *Political Behavior*, 39(3), 629-649.

Polit Behav (2017) 39:629–649
DOI 10.1007/s11109-016-9373-5



ORIGINAL PAPER

Tweetment Effects on the Tweeted: Experimentally Reducing Racist Harassment

Kevin Munger¹

Published online: 11 November 2016
© Springer Science+Business Media New York 2016

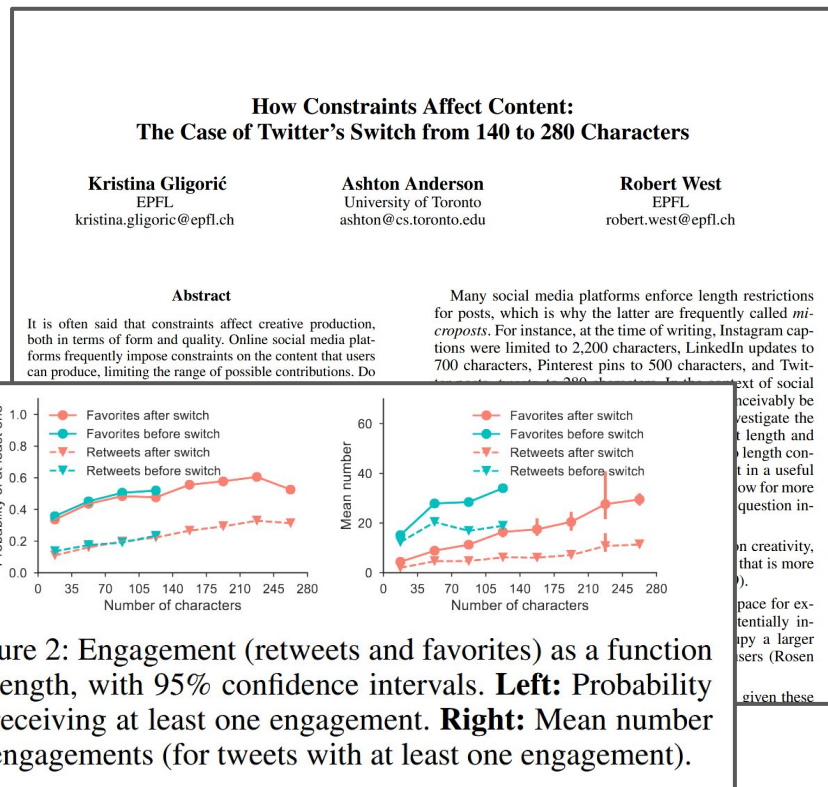
Abstract I conduct an experiment which examines the impact of group norm promotion and social sanctioning on racist online harassment. Racist online harassment de-mobilizes the minorities it targets, and the open, unopposed expression of racism in a public forum can legitimize racist viewpoints and prime ethnocentrism. I employ an intervention designed to reduce the use of anti-black racist slurs by white men on Twitter. I collect a sample of Twitter users who have harassed other users and use accounts I control (“bots”) to sanction the harassers. By varying the identity of the bots between in-group (white man) and out-group (black man) and by varying the number of Twitter followers each bot has, I find that subjects who were sanctioned by a high-follower white male significantly reduced their use of a racist slur. This paper extends findings from lab experiments to a naturalistic setting using an objective, behavioral outcome measure and a continuous 2-month data collection period. This represents an advance in the study of prejudiced behavior.

Keywords Online harassment · Social media · Randomized field experiment · Social identity

‘Natural’ Experiments on social media platforms

- Common sources on social media:
 - Sudden platform policy changes (e.g., bans, feed redesigns, deplatforming)
 - External shocks (news events, lockdowns, elections)
 - Staggered rollouts of features across users or regions
- Example: Do character limits on social media impact how people tweet?

Gligorić, Kristina, Ashton Anderson, and Robert West. "[How constraints affect content: The case of Twitter's switch from 140 to 280 characters.](#)" *Proceedings of the International AAAI Conference on Web and Social Media*.



Related: Event-centred studies

- Use a real-world event as the "treatment" — a natural experiment
 - Did something change after event X that we wouldn't expect otherwise?
- Common designs: Difference-in-differences (DiD) or Regression Discontinuity
- Key assumption: the event is plausibly **exogenous** to the outcome
 - Exogenous: outside the model, i.e., not associated with the treatment
 - Opposite of endogenous

Event-centred study: example

- RQ: Does deplatforming hate organizations actually reduce hate on the platform?
 - Treatment: Facebook removes leadership of designated hate organizations
 - Outcome: hate-content consumption + production by the audience
- Design: difference-in-differences across six staggered organization-level events
 - Treated: members of the hate organization's audience
 - Control: similar users not exposed to the disruption
 - Compare changes in hateful content before vs after the event
- Finding: deplatforming reduced consumption and production of hateful content on average
 - Some short-term backlash among the closest core members

Thomas, D. R., & Wahedi, L. A. (2023). [Disrupting hate: The effect of deplatforming hate organizations on their online audience](#). *Proceedings of the National Academy of Sciences*, 120(24), e2214080120.

PNAS

RESEARCH ARTICLE | POLITICAL SCIENCES

OPEN ACCESS



Disrupting hate: The effect of deplatforming hate organizations on their online audience

Daniel Robert Thomas* and Laila A. Wahedi*

Edited by Timothy Wilson, University of Virginia, Charlottesville, VA; received August 17, 2022; accepted January 20, 2023

How does removing the leadership of online hate organizations from online platforms change behavior in their target audience? We study the effects of six network disruptions of designated and banned hate-based organizations on Facebook, in which known members of the organizations were removed from the platform, by examining the online engagements of the audience of the organization. Using a differences-in-differences approach, we show that on average the network disruptions reduced the consumption and production of hateful content, along with engagement within the network among periphery members. Members of the audience closest to the core members exhibit signs of backlash in the short term, but reduce their engagement within the network and with hateful content over time. The results suggest that strategies of targeted removals, such as leadership removal and network degradation efforts, can reduce the ability of hate organizations to successfully operate online.

hate speech | deplatforming | social networks | leadership removal

Hate-based, terrorist, and criminal organizations attempt to use online platforms to spread their ideology, recruit new members, and coordinate existing members (1). They attempt to leverage online networks because of their ability to spread information quickly and widely. To counter this, social media platforms reduce the ability of such organizations to organize on their platforms, often by removing the members, or “deplatforming” them.¹ To what extent does deplatforming successfully mitigate harms? To answer this question, we employ a differences-in-differences design, taking advantage of the staggered timing of six exogenous organization-level deplatforming events, and granular daily user-level observations.² We show a strong overall negative effect: Deplatforming reduces hate on the platform.

We study the effects of “strategic network disruptions” (SNDs), a method of deplatforming in which identifiable core members of a hate-based organization are removed from the platform all at once, eliminating the online leadership of the organization. The goal of this approach is to disrupt the operation of the organization by removing key actors at the same time in order to make it more difficult for them to rebuild their audiences and regroup. If the approach is successful, then the organization’s online target audience should be exposed to less radicalizing content and hate, and decrease their production of it. We study the effects of the disruptions on this target audience.

We find that disruptions create a healthier platform on average: Members of the audience of the hate-based organizations reduce their consumption and production of hateful content, and engage less with other audience members. However, these average results mask substantial heterogeneity between audience members who engaged most with the core prior to disruption, and those least engaged. While the least engaged subgroup reduces its engagement with hateful content and with other audience members following the disruptions, the most engaged subgroup exhibits signs of backlash, increasing its consumption and production of hateful content,³ and engaging more with other audience

Significance

How does removing the leadership of online hate organizations from online platforms change behavior in their target audience? We study the effects of six network disruptions of designated and banned hate-based organizations on Facebook, in which known members of the organizations were removed from the platform, by examining the online engagements of the audience of the organization. We show that on average, the network disruptions reduced the consumption and production of hateful content, along with engagement within the network among audience members. The results suggest that strategies of targeted removals, such as leadership removal and network degradation efforts, can reduce the ability of hate organizations to successfully operate online.

Author affiliations: *Meta, Menlo Park, CA 94025

Author contributions: D.R.T. and L.A.W. designed research; performed research; analyzed data; and wrote

¹Organizations that proclaim a violent mission, engage in violence, and represent ideologies that promote hate are

Matching example: Conspiracy Theorists on Reddit

- RQ: why people are drawn to conspiracy communities
- Platform: Reddit
- Match future conspiracists (FC) with non-conspiracists (CC)
 - Reddit joining time
 - contribution volumes across different time spans
 - subreddits they contribute to
- Compare social factors

Phadke, S., Samory, M., & Mitra, T. (2021). [What makes people join conspiracy communities? Role of social factors in conspiracy engagement.](#) *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW3), 1-30.

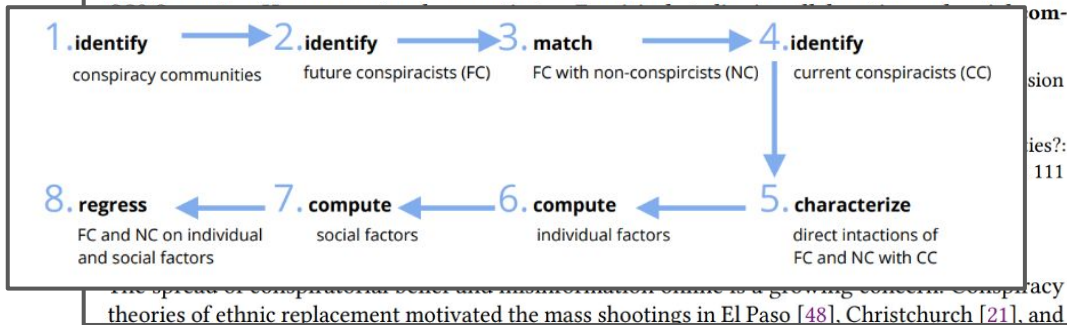
What Makes People Join Conspiracy Communities?: Role of Social Factors in Conspiracy Engagement

SHRUTI PHADKE, University of Washington, USA

MATTIA SAMORY, GESIS, Germany

TANUSHREE MITRA, University of Washington, USA

Widespread conspiracy theories, like those motivating anti-vaccination attitudes or climate change denial, propel collective action and bear society-wide consequences. Yet, empirical research has largely studied conspiracy theory adoption as an individual pursuit, rather than as a socially mediated process. What makes users join communities endorsing and spreading conspiracy theories? We leverage longitudinal data from 56 conspiracy communities on Reddit to compare individual and social factors determining which users join the communities. Using a quasi-experimental approach, we first identify 30K *future conspiracists*—(FC) and 30K matched *non-conspiracists*—(NC). We then provide empirical evidence of importance of social factors across six dimensions relative to the individual factors by analyzing 6 million Reddit comments and posts. Specifically in social factors, we find that dyadic interactions with members of the conspiracy communities and marginalization outside of the conspiracy communities, are the most important social precursors to conspiracy joining—even outperforming individual factor baselines. Our results offer quantitative backing to understand social processes and echo chamber effects in conspiratorial engagement, with important implications for democratic institutions and online communities.



Pitfalls to watch out for when doing Causal Inference

Carefully thinking about confounders

- List your suspected confounders BEFORE looking at the data
 - User-level: age, prior activity, network size, language
 - Content-level: topic, length, time-of-day
 - Platform-level: algorithmic exposure, recommendations
- You can only adjust for what you measure
 - Unmeasured confounders bias results
 - Within-study comparisons on Twitter found that even careful matching often fails to recover the experimental effect
- Be clear in limitations about potential unobserved confounders, e.g., views

Causal language: say what you actually mean

Match the strength of your verbs to the strength of your design.

DESCRIPTIVE / CORRELATIONAL

is associated with

correlates with

co-occurs with

predicts

users who post X also post Y

CAUSAL

causes

leads to

results in

increases / decreases

posting X makes users post more Y

Reviewers, journalists, and readers will (and should) hold you to the verbs you use. As will the evaluators of your course projects.

If your design is correlational, your title shouldn't say "X causes Y".

Summary and Next Week

- This session we learned:
 - Causal vs. descriptive questions
 - Techniques for doing causal inference
 - Experiments
 - Observational setups — natural experiments and matching-based approaches
 - Best practices for causal inference with social media
- Next week:
 - Ethical Considerations when working with social media data
- Readings:
 - Kramer, Adam DI, Jamie E. Guillory, and Jeffrey T. Hancock. "[Experimental evidence of massive-scale emotional contagion through social networks.](#)" Proceedings of the National Academy of Sciences
 - Hudson, James M., and Amy Bruckman. "[“Go away”: Participant objections to being studied and the ethics of chatroom research.](#)" The information society 20.2 (2004): 127-139. 2.

Questions?

Additional Resources

- Matheus Facure Alves: [Causal Inference for the Brave and True](#)
- Amit Sharma's [Causal Inference Tutorial](#)